Announcements

- Homework 3 is ongoing, due this Thursday
- Quiz 5 today (on Topic 6)
- Midterm 2 is April 16th

Topic 7: Binary Search Trees

By: Professor Lynam

General Trees

- Definition: *General Tree*
- *A general tree T is empty or is a finite set of Nodes N such that there is a specially-designated root node r* ∈ N with the remaining N – r nodes partitioned into $m \geq 0$ *disjoint subtrees of T*
- Binary Search Trees ⊆ Binary Trees ⊆ General Trees
- Note that this is a recursive definition, meaning you can have sub-trees, and trees being forming by attaching existing trees
- A node in a general tree may have any non-negative quantity of "children"

Binary Trees

- Definition: *Binary Tree*
- *A binary tree is a general tree such that* $0 \leq m \leq 2$.

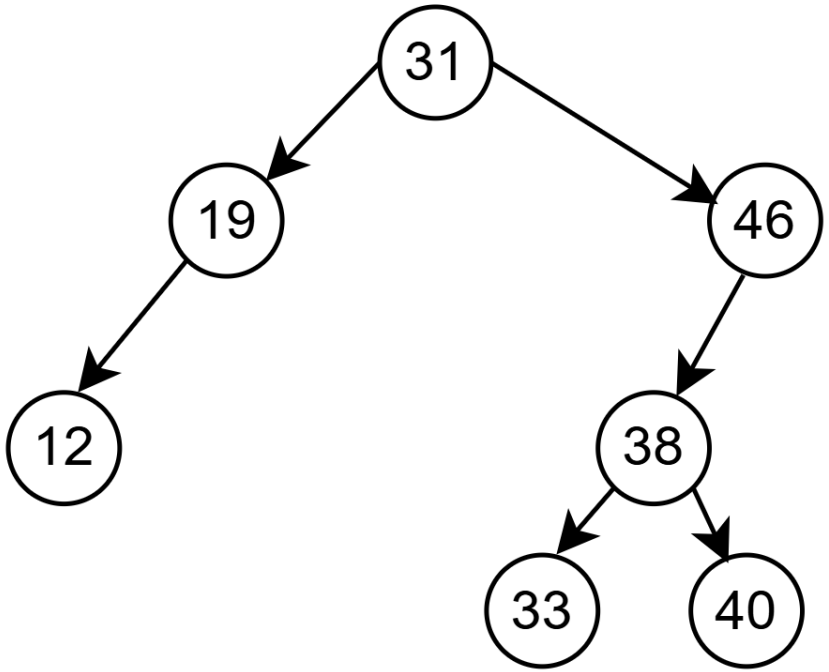
- We call the node "higher" in the tree the parent node, and assuming it has two children, it has a "left" and "right" child node
- Example applications of binary trees include:
- Binary Space Partitioning (BSP) in video games
- Huffman Coding (lossless data compression)
- Heaps (priority queues)
- Expression Trees (intermediate expression representation)
- And last, but not least…

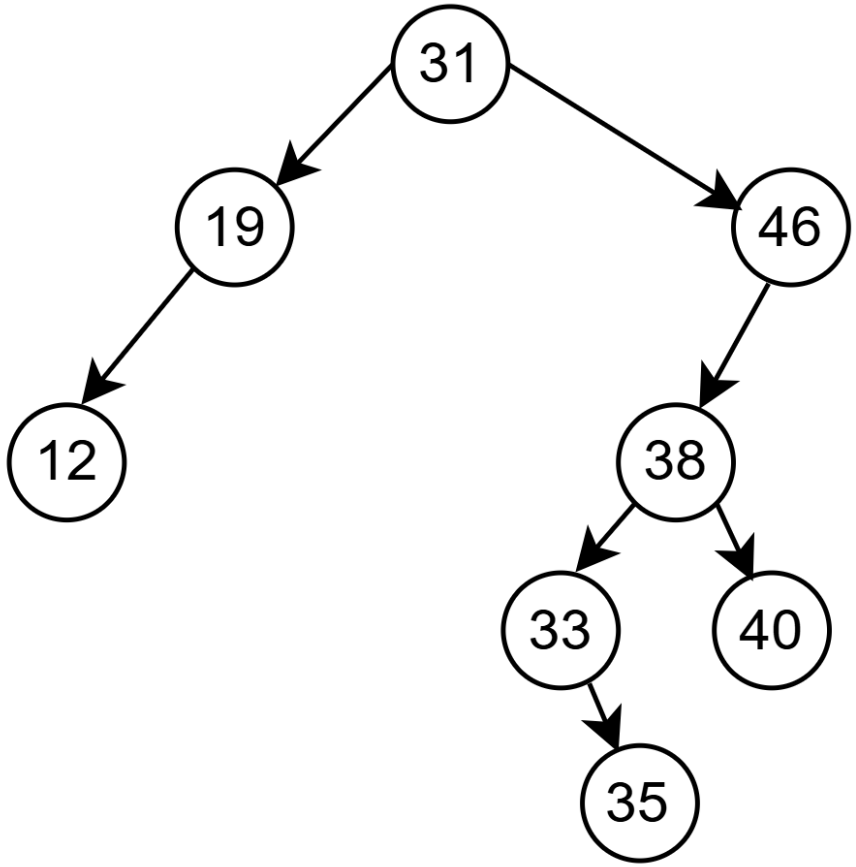
Binary Search Trees (BSTs)

- Definition: *Binary Search Tree*
- *A binary search tree (BST) is a binary tree in which any key value held in a node is greater than any key in that node's left subtree and is less than any key in its right subtree.*

- What about equal key values, you might ask?
- We can extend the definition to store ties on one side or the other (but not both!) of existing matches. (However, in many BST applications, data values are unique.)

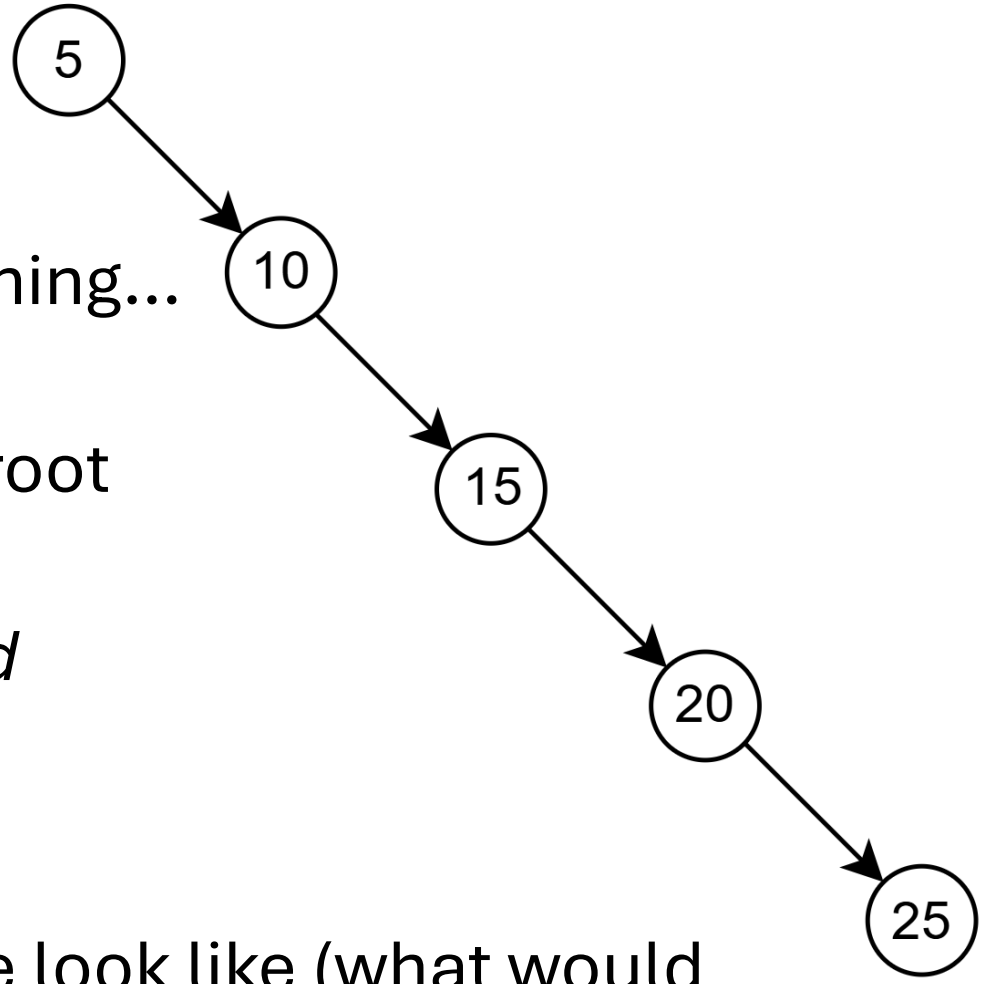
BST Insertions





- Let's insert 35 into this BST. What are the steps?
- Just search for 35, and insert into the fringe where the search fails

BST Searching



- A key advantage of BSTs is efficient searching… let's see how efficient it actually is
- Best case: O(1), when the target is in the root node
- Worst case: O(n), when the tree is *skewed*
- What does a skewed BST look like?
- A Linked List (searching an LL is O(n) too)
- What would the opposite of a skewed tree look like (what would our *ideal* tree look like?)?
- A "full" or "complete" tree, with each level filled
- Average case: depends on the tree shape! Could be O(*h*) to O(n/2)

BST Searching

- Definition: *Complete Tree*
- *A complete tree of height h, where the root is h = 0, has all positions on all levels filled with data.*

- So, what's the capacity of a height *h* binary tree?

Height	Capacity
0	1
1	3 (1 + 2)
2	7 (3 + 4)
3	15 (7 + 8)
…	…
h	$2^{h+1} - 1 (= n)$

BST Searching

- The height of the BST correlates with the number of comparsions necessary for searching… So what is *h* in terms of *n* (the quantity of data)?
- $2^{h+1} - 1 = n$
- $2^{h+1} = n + 1$
- $\log_2 2^{h+1} = \log_2(n + 1)$
- $h = \log_2(n + 1) - 1$
- So h is O($\log_2 n$)
- But most BST aren't complete. Suppose one is just half-full… All we'd have to do is replace n with n/2 in our equation. We'd still get a $\log_2 n$ term

Tree Sort

- The idea for tree sort is that an inorder traversal of a BST visits the keys in ascending order
- The algorithm outline is simple:

1. Build a BST
2. ‘Inorderly’ traverse the BST

Tree Sort

- Analysis time! What's our best case scenario?
- The tree will end up being complete!

1. Build a BST: n insertions at O($\log_2 n$) each -> O($\mathrm{n} * \log_2 n$)
2. 'Inorderly' traverse the BST: T(n) = T(n/2) + 1 + T(n/2)… which is?

- O(n) by the master theorem! So overall, O($\mathrm{n} * \log_2 n$)
- Worst case? We get a completely skewed tree!

1. Build a BST: 1 + 2 + 3 + … + n -> O(n^2)
2. Traverse: still O(n)

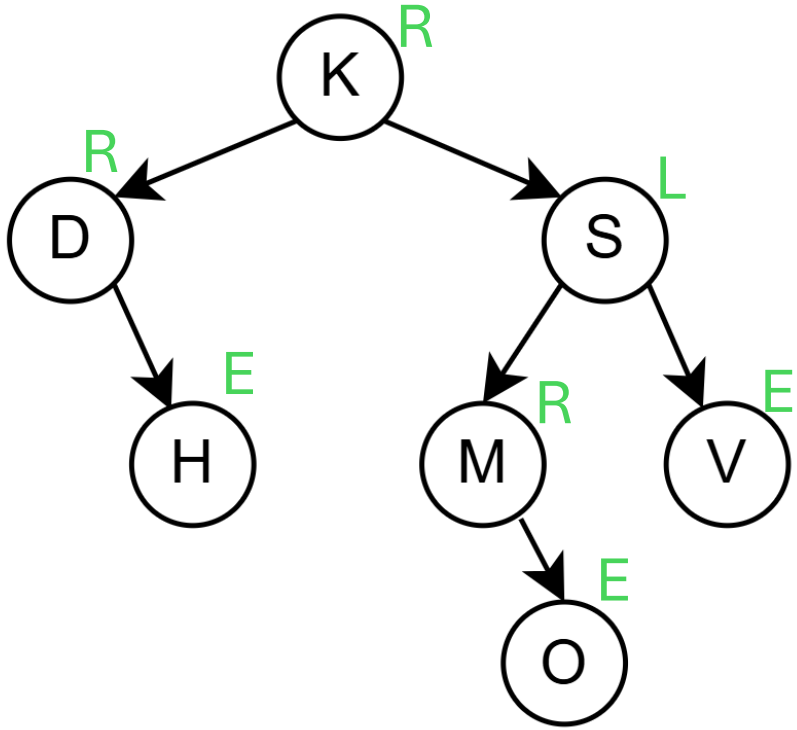
- Overall, O(n^2)
- Average case? Same as best case: O($\mathrm{n} * \log_2 n$)

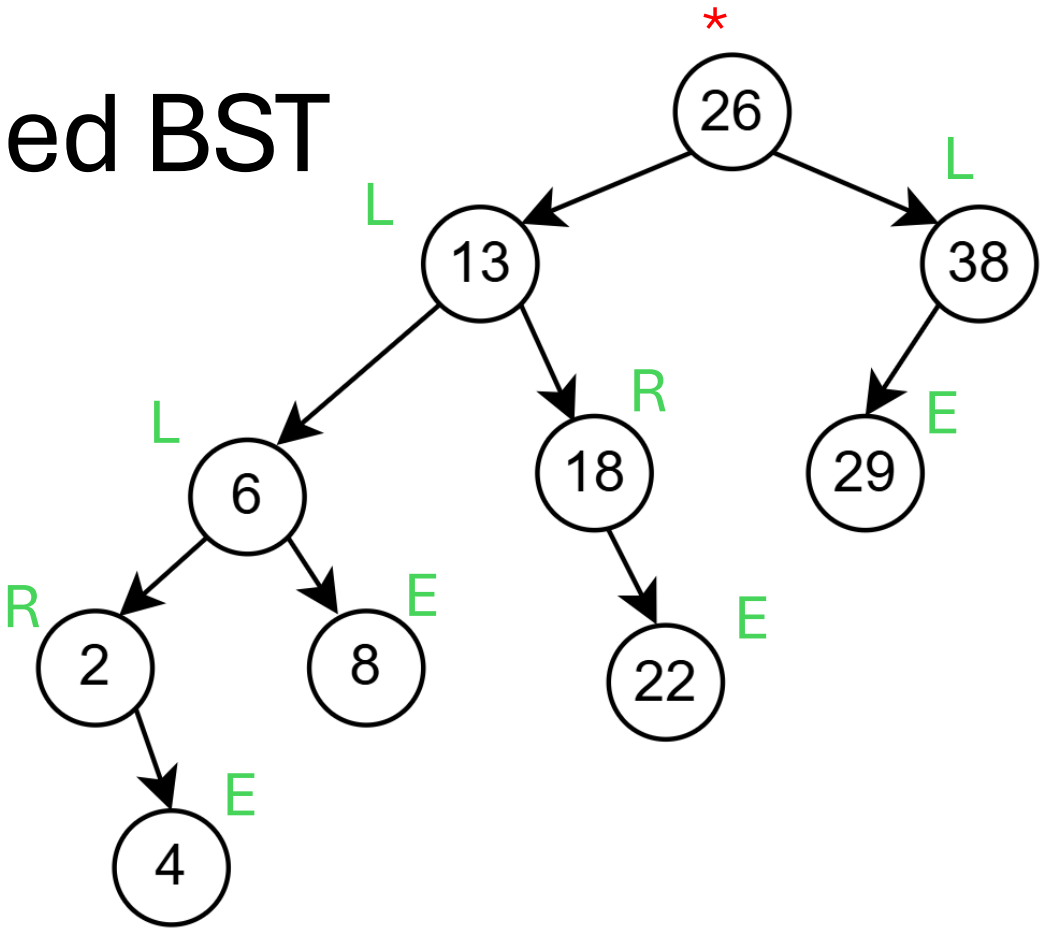
AVL Trees: A Height-Balanced BST

- Definition: *AVL Tree*
- *An AVL tree is a BST satisfying this balance condition: For each node in the tree, the height of its left subtree differs from the height of its right subtree by no more than one.*

- The name is for "Adelson-Velsky and Landis," who invented them in 1962
- If this is a height-balanced BST, do weight-balanced trees exist?
- Yes! Huffman coding trees are one such example.

AVL Trees: A Height-Balanced BST





- E: subtree is evenly balanced
- L: subtree is left-heavy by one level
- R: subtree is right-heavy by one level
- *: marks the *pivot*, the root of the deepest unbalanced subtree

AVL Trees Insertions

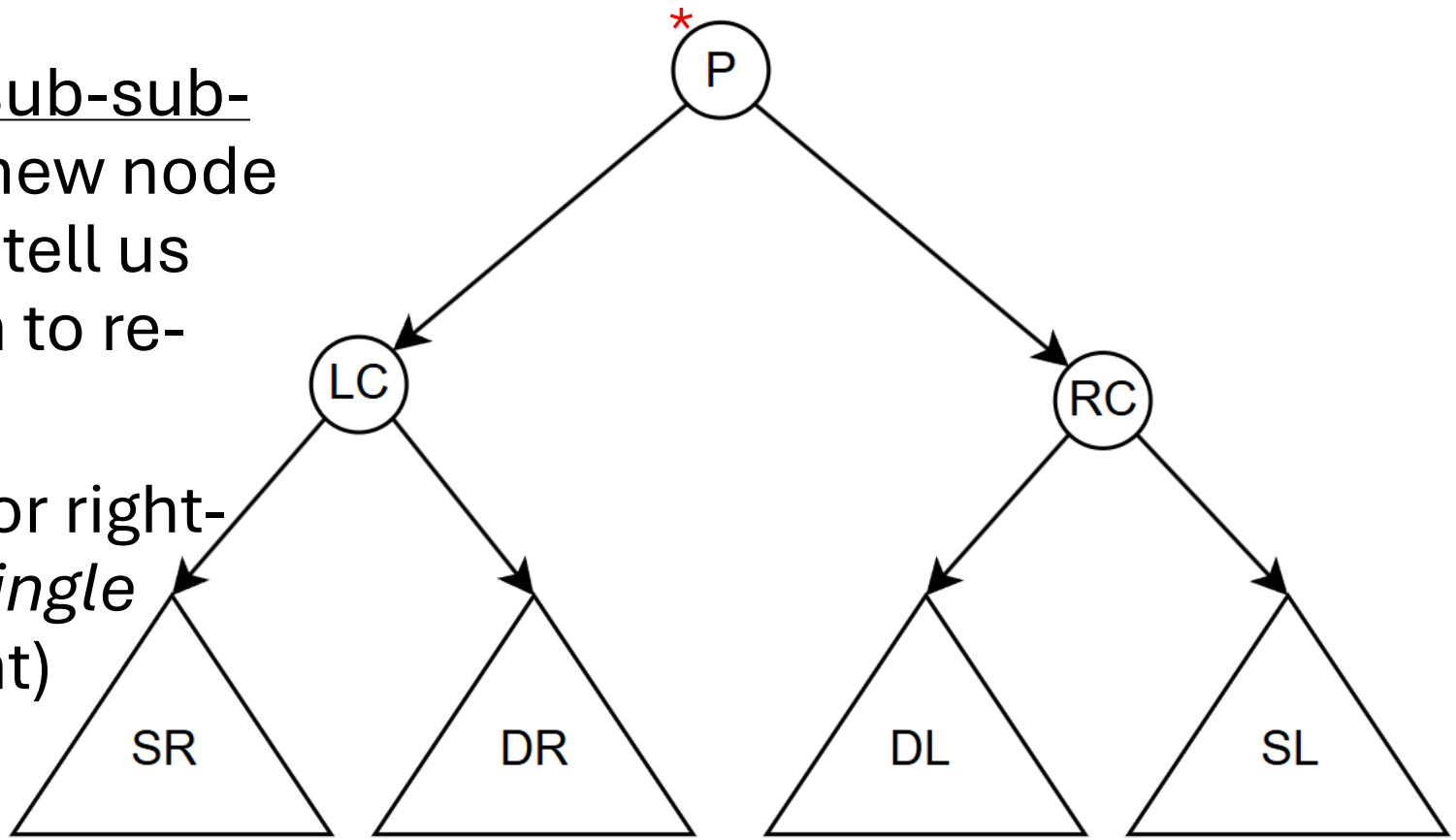
- Just two steps!

1. Perform a normal BST insertion.
2. (if necessary) Rebalance the tree.

- Done! Easy.

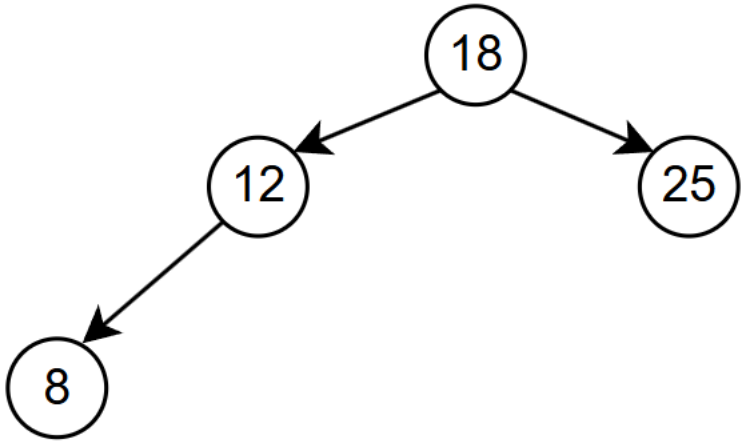
AVL Trees Rebalancing

- We need to know which <u>sub-sub-tree</u> of the pivot was the new node inserted. The answer will tell us which *rotation* to perform to re-balance the tree
- Insertions to the left-left or right-right subtrees require a *single rotation* (either left or right)
- Insertions to the left-right or right-left subtrees require a *double rotation* (again, either left or right)



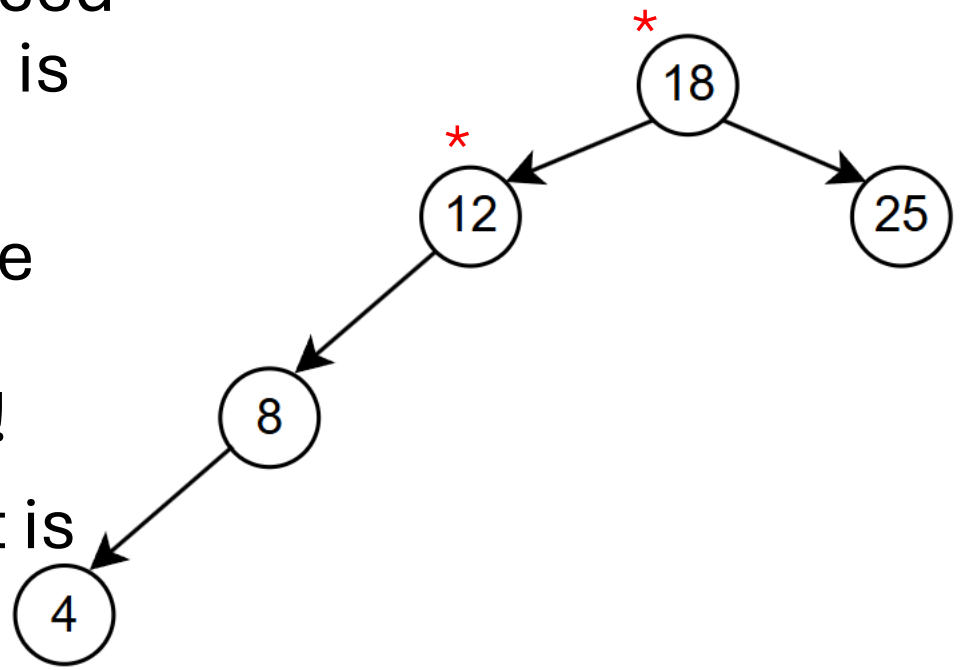
AVL Trees Insertions

- Insert 4 into this AVL tree:
- First, is it an AVL tree? What is each nodes balance?



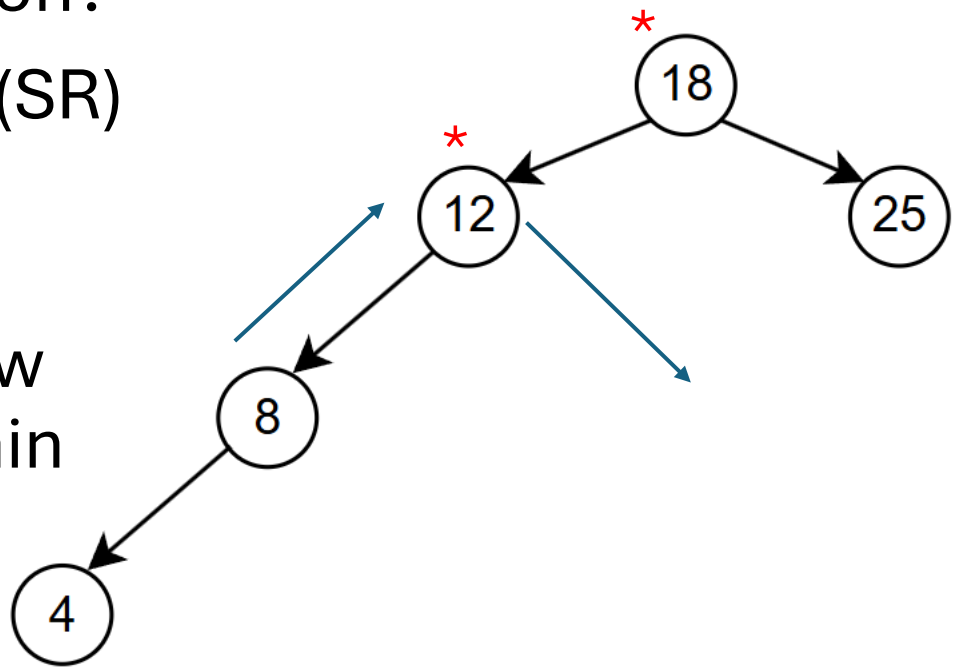
AVL Trees Insertions

- Insert 4 into this AVL tree:
- Is it unbalanced now? Where is the pivot?
- 12 and 18 are both unbalanced!
- But the pivot is the *deepest* unbalanced subtree root



AVL Trees Insertions

- Insert 4 into this AVL tree:
- Which rotation?
- Single Right (SR)
- Imagine a snowplow pushing snow up a mountain



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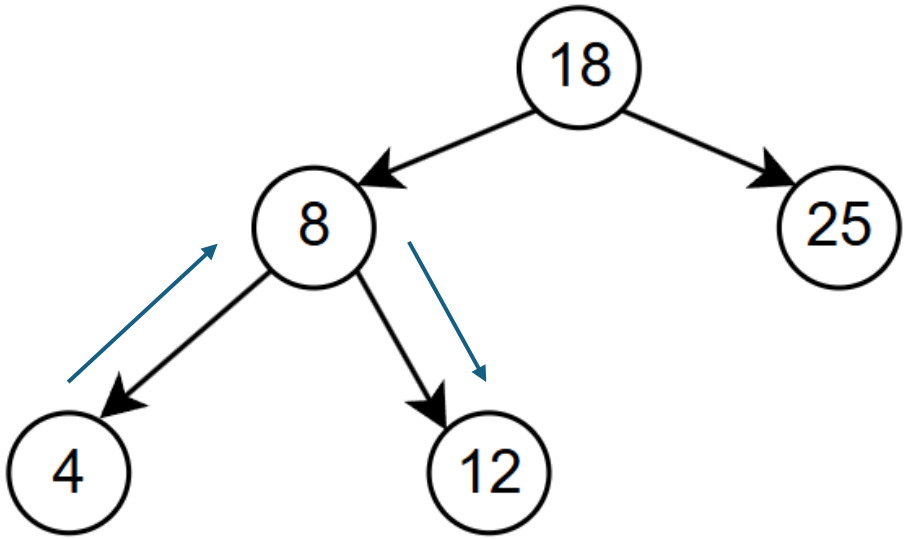
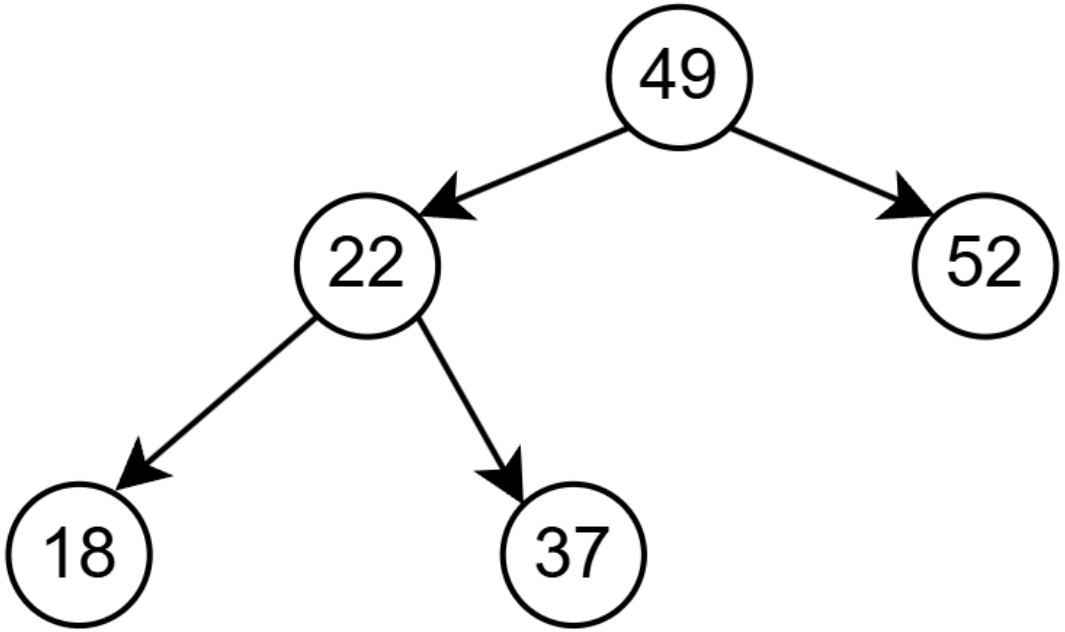


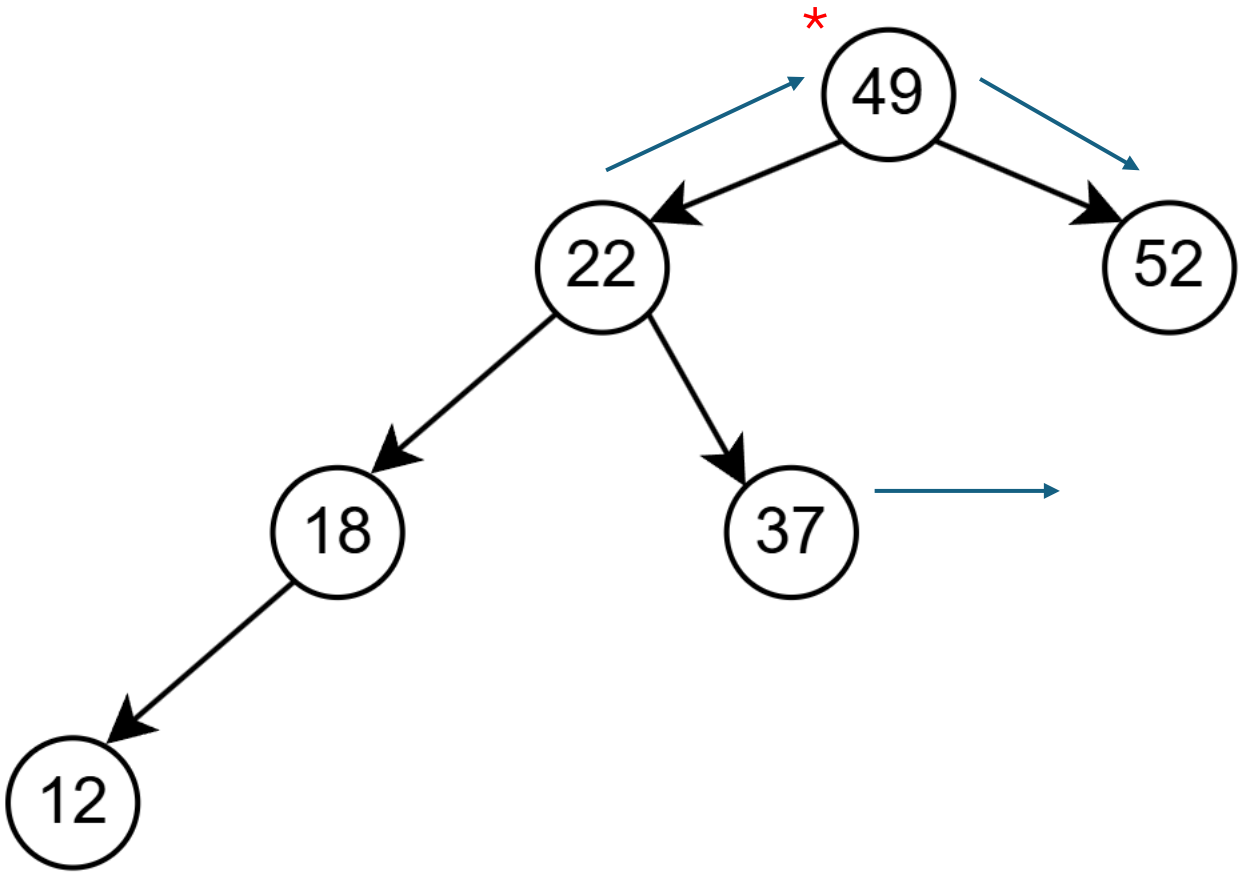
AVL Trees Insertions

- Insert 12 into this AVL tree:
- First, is it an AVL tree? What is the balance of each node?
- Where would 12 go? Would it cause it to become unbalanced?



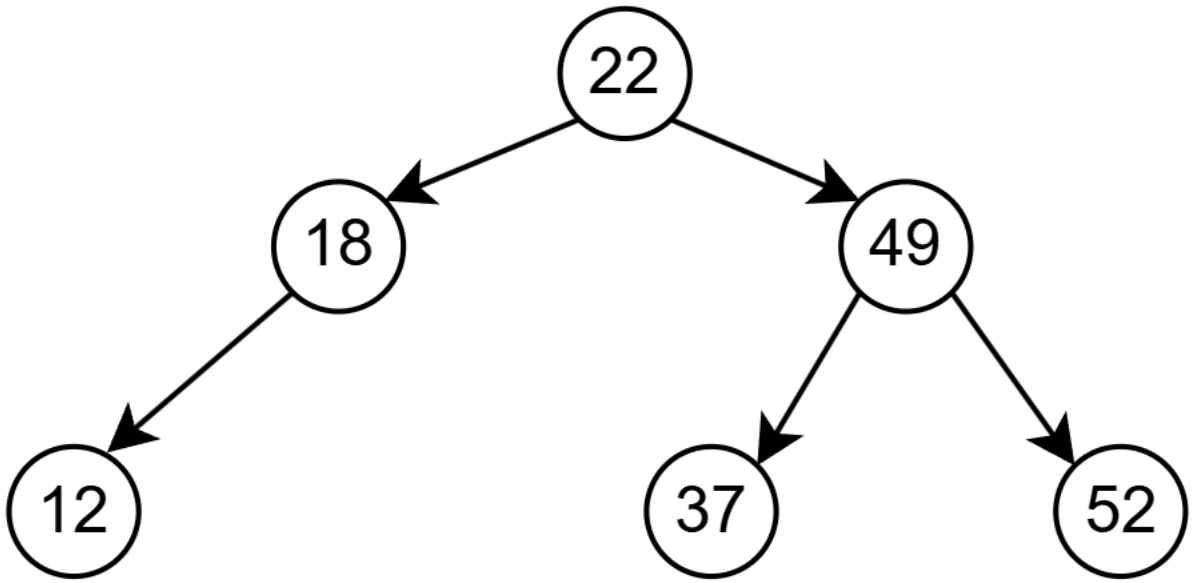
AVL Trees Insertions

- Insert 12 into this AVL tree:
- What would the pivot be?
  What would the rotation be?



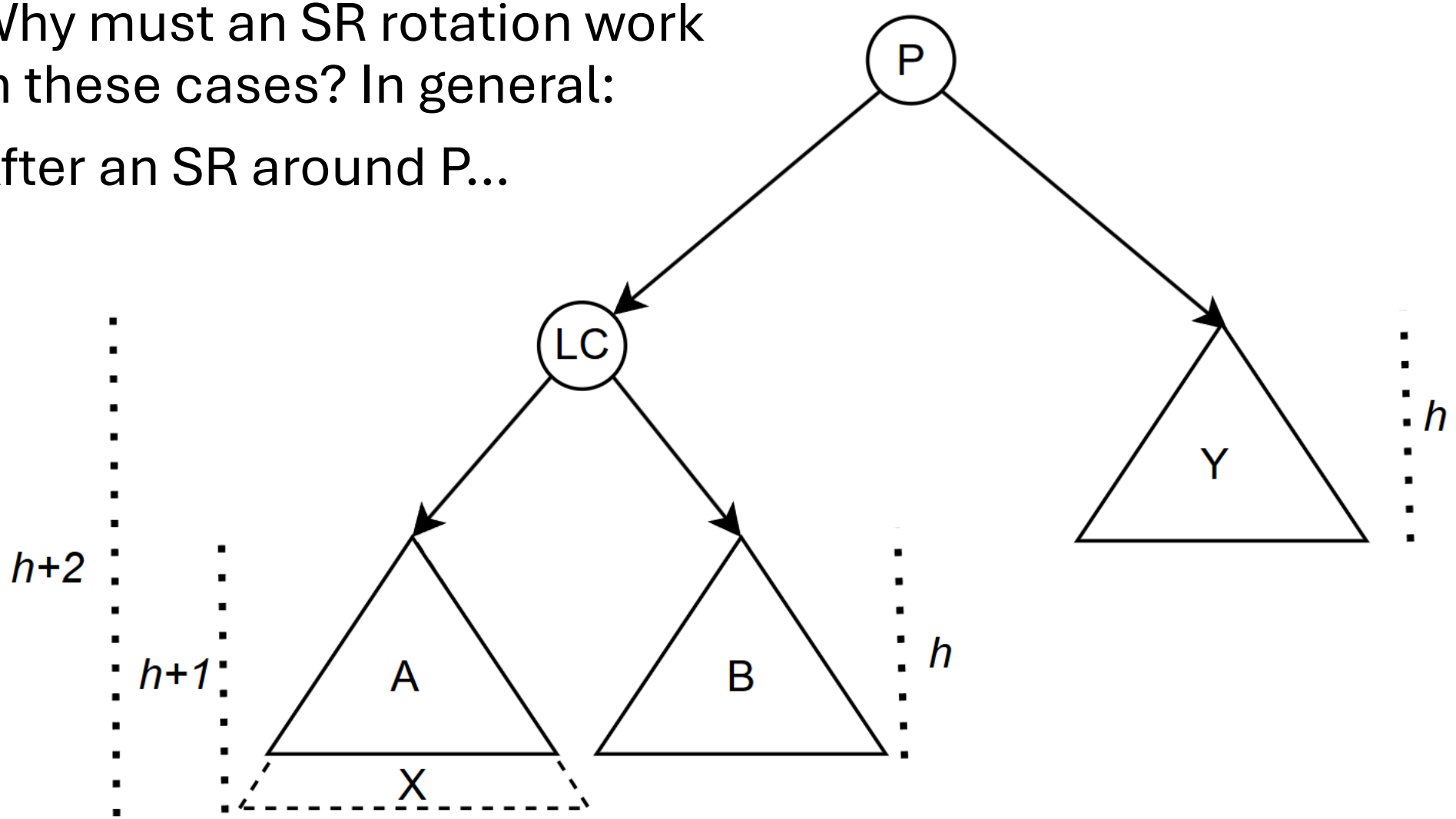
AVL Trees Insertions

- Insert 12 into this AVL tree:
- Another SR, with the right subtree of 22 becoming the left subtree of 49
- There will always be a place for every subtree in the rotated tree



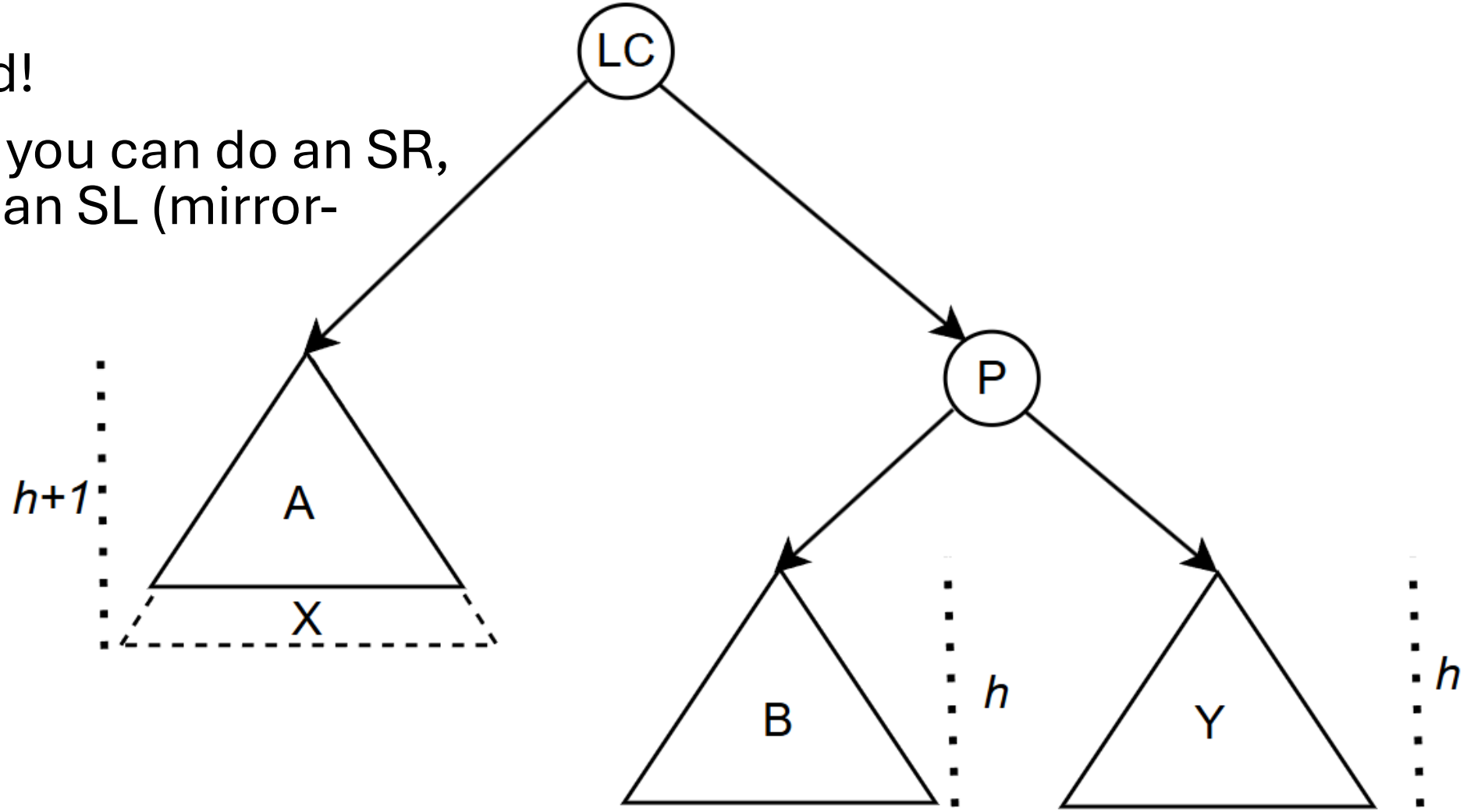
AVL Trees Single Rotations

- Why must an SR rotation work in these cases? In general:
- After an SR around P…



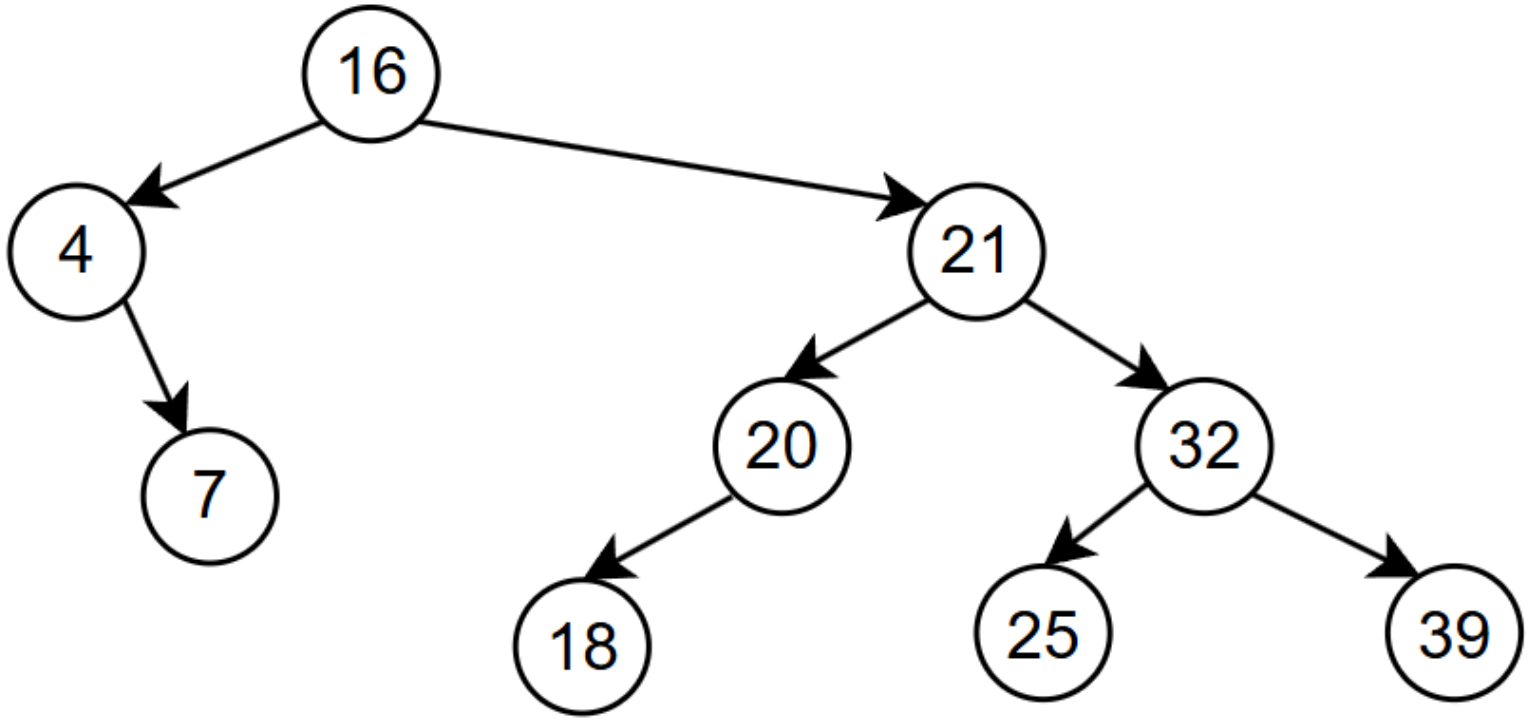
AVL Trees Single Rotations

- Rebalanced!
- Note that if you can do an SR, you can do an SL (mirror-image!)



AVL Trees Insertions

- Insert 36 to this AVL tree.



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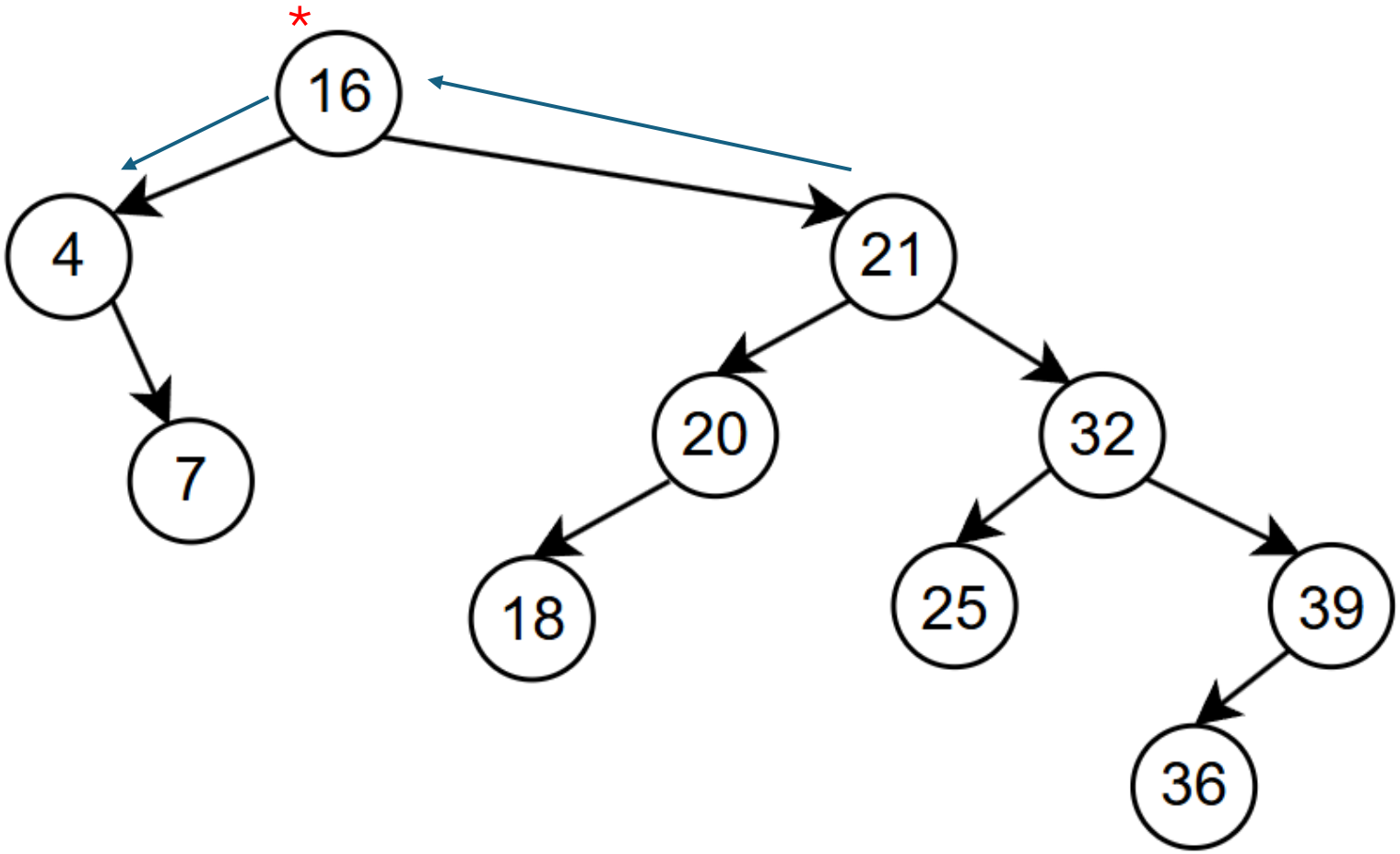

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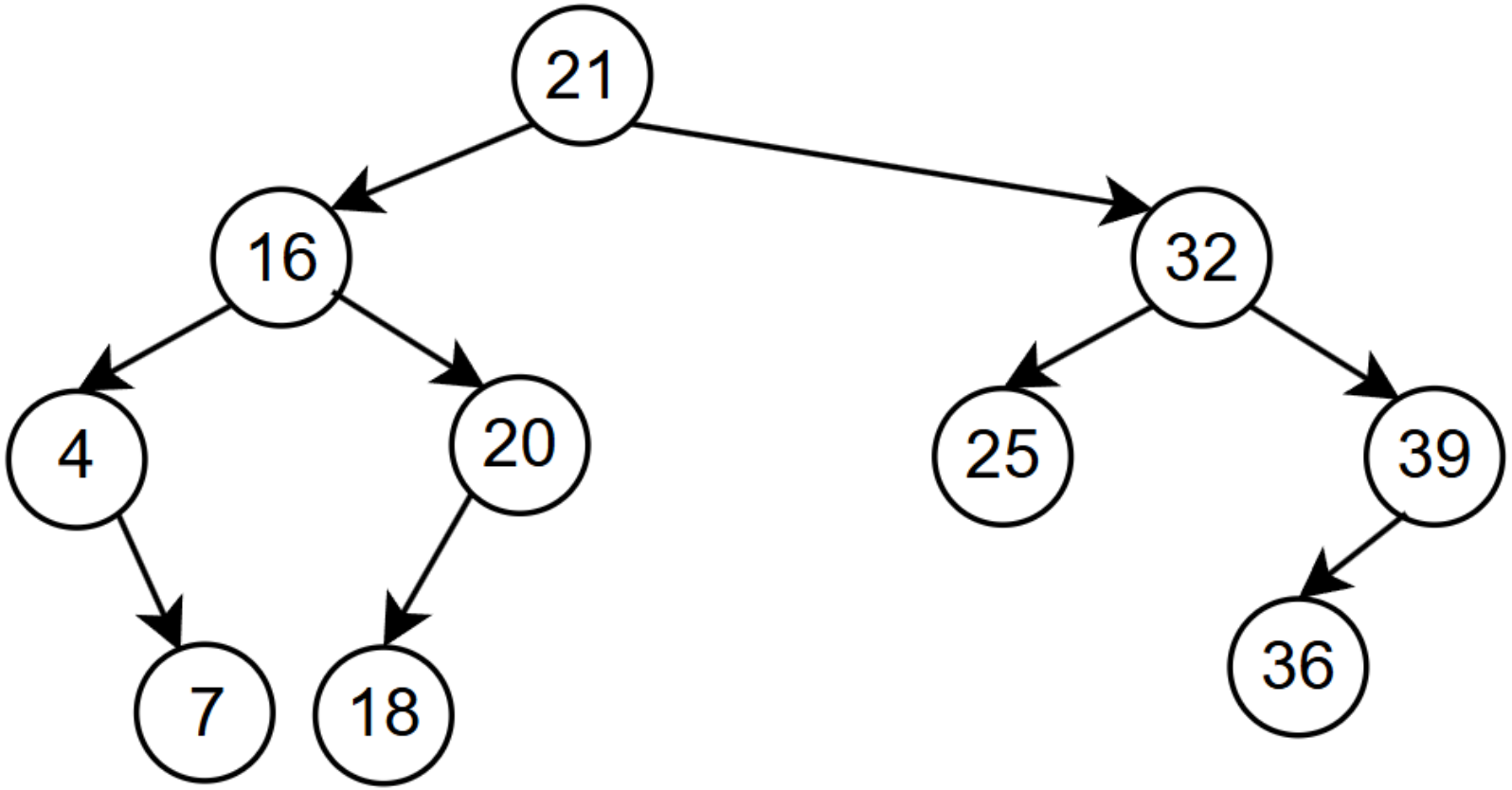
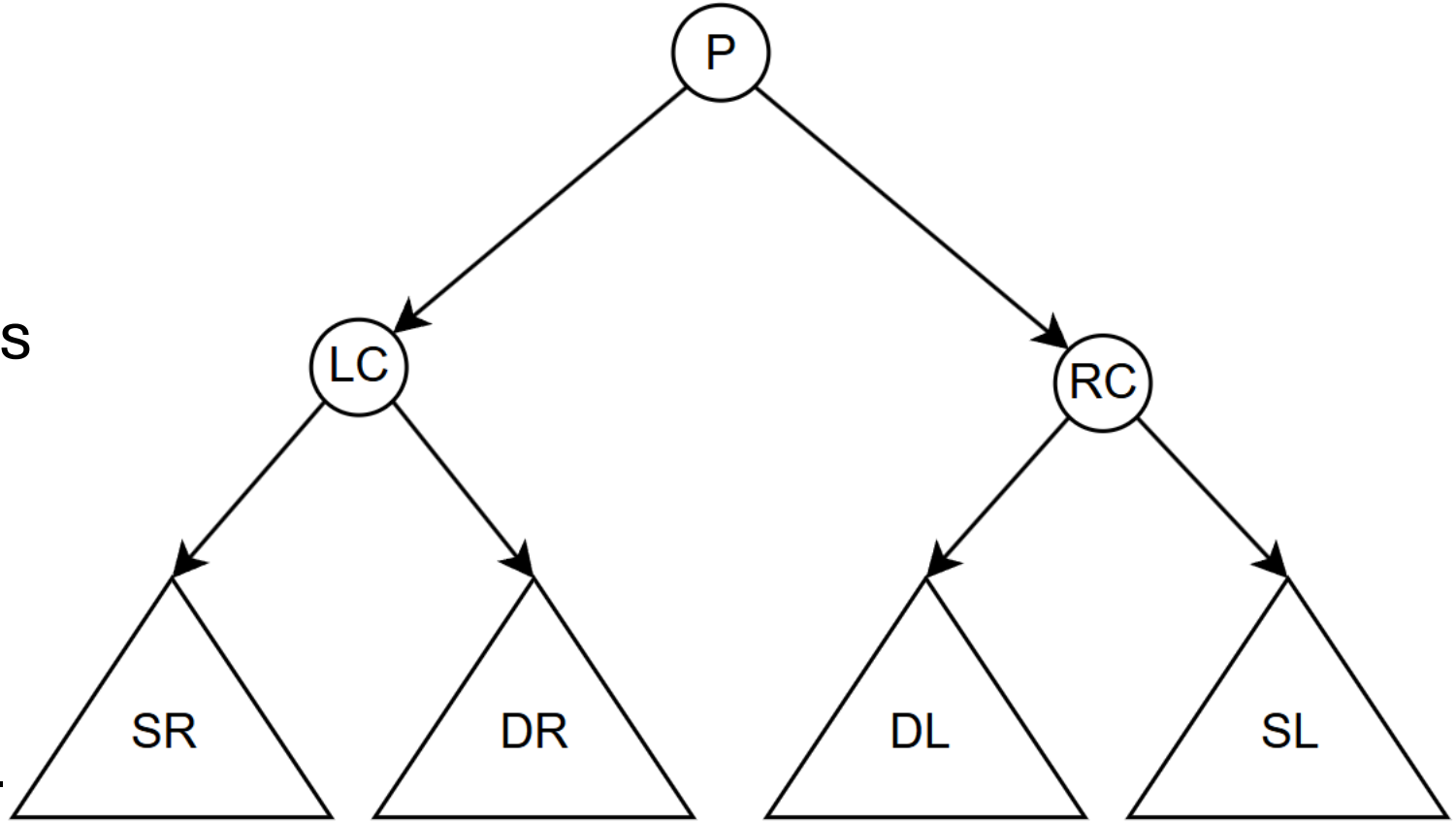


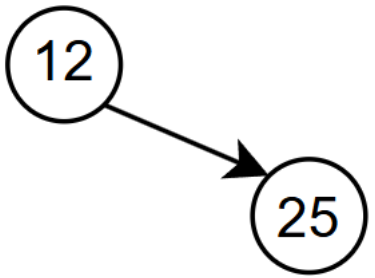
AVL Trees Double-Rotations

- Recall our pivot diagram:
- When insertions add depth to the inner sub-sub-trees, single rotations won't rebalance the tree
- We'll need *two* single rotations… aka a *double* rotation
- The result is the sub-sub-tree's root replacing the pivot (and that root is where the rotations start)



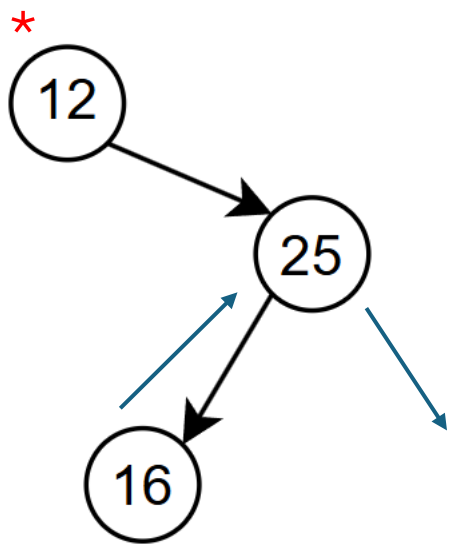
AVL Trees Double Rotations

- Insert 16 to this AVL tree.



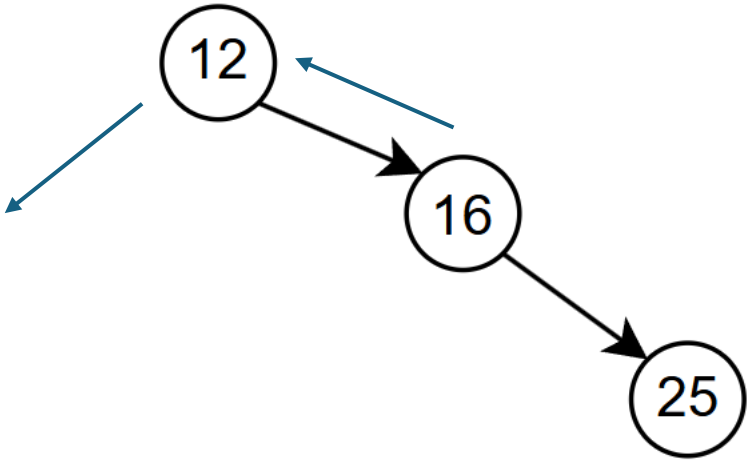
AVL Trees Double Rotations

- Insert 16 to this AVL tree.
- The goal is to get the sub-sub-tree's root two levels higher. Just follow the tree's structure to know which rotations to perform (and in which order!)
- Also note: one single rotation can't rebalance the tree: try it!



AVL Trees Double Rotations

- Insert 16 to this AVL tree.
- Start with an SR
- End with an SL



AVL Trees Double Rotations

- Insert 16 to this AVL tree.
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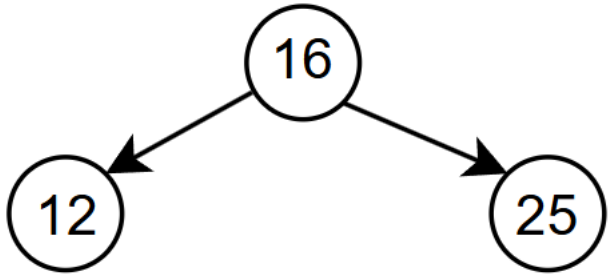
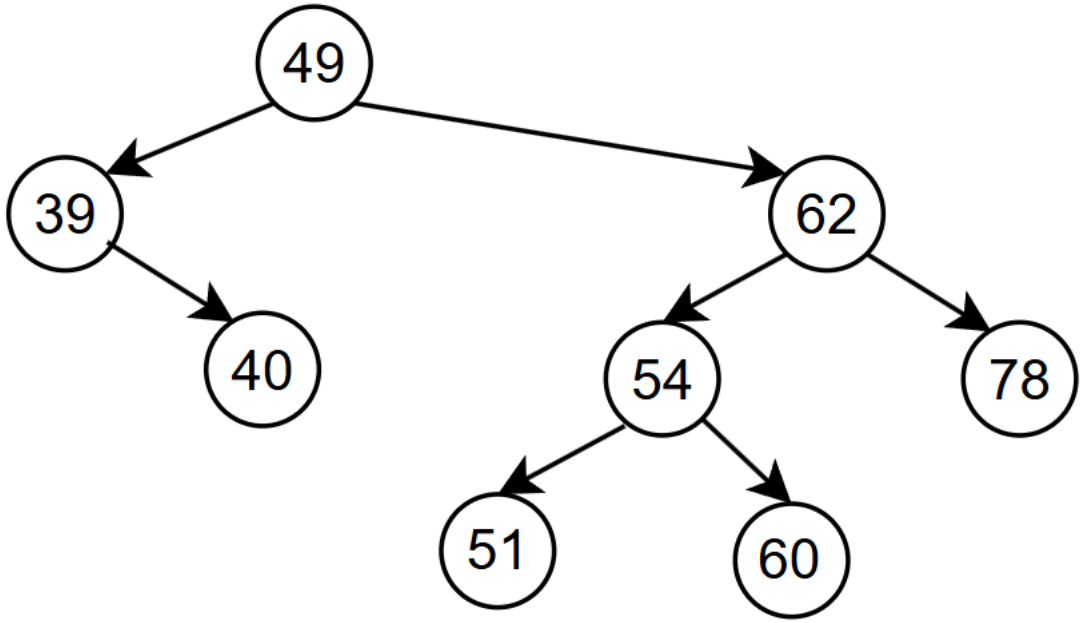


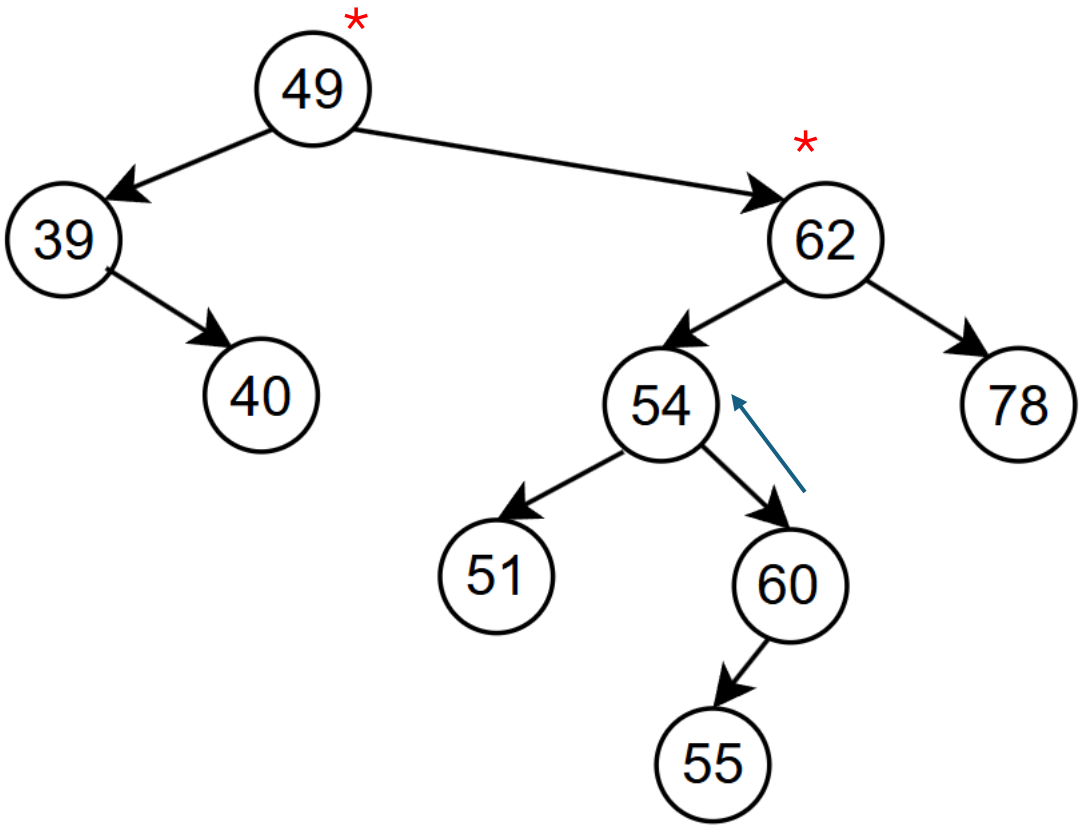
AVL Trees Double Rotations

- Insert 55 to this AVL tree.



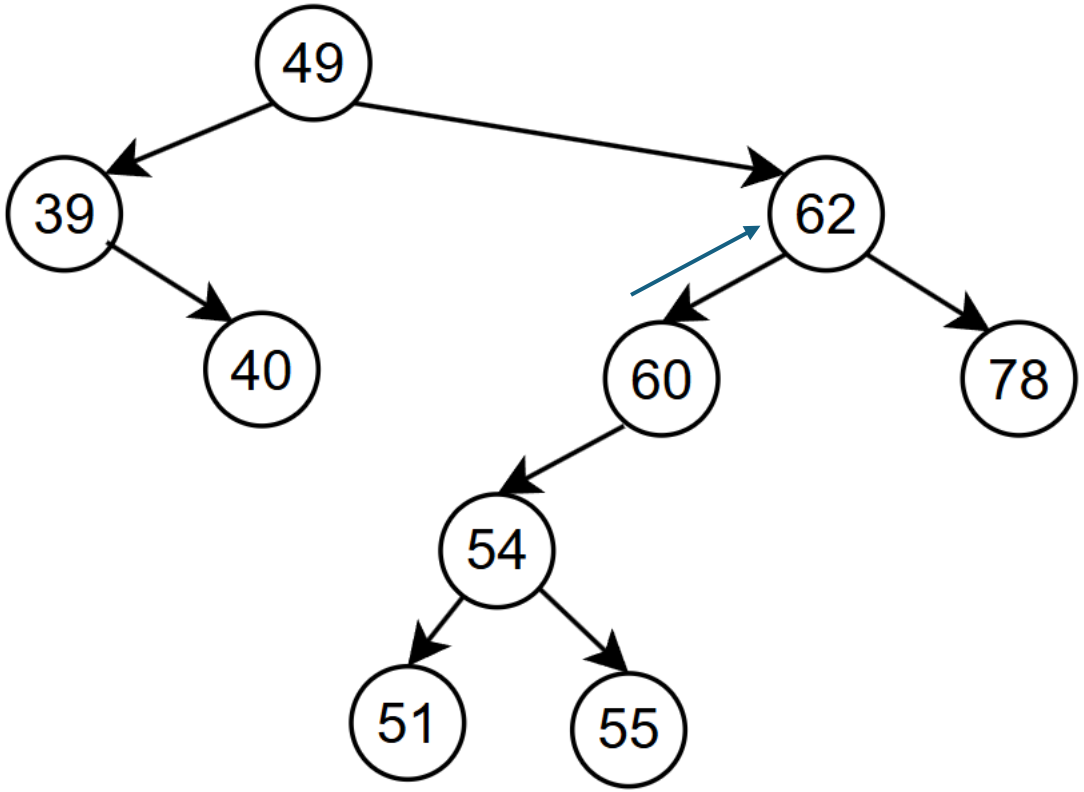
AVL Trees Double Rotations

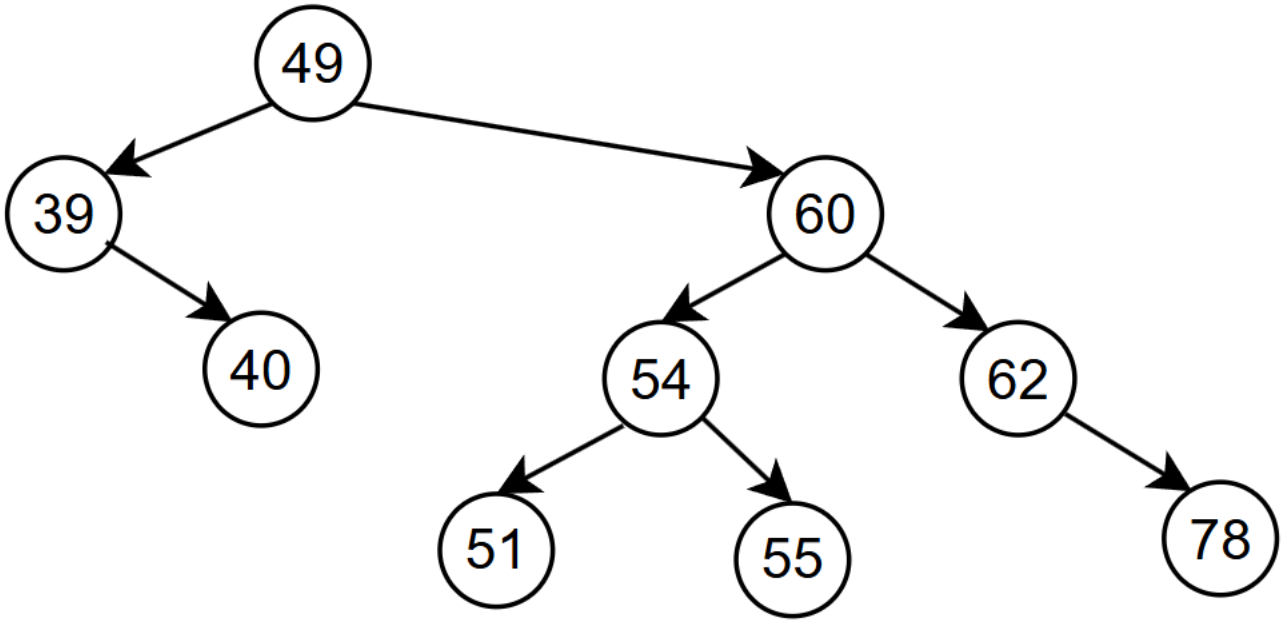
- Insert 55 to this AVL tree.
- First, SL



AVL Trees Double Rotations

- Insert 55 to this AVL tree.
- Now, SR



AVL Trees Double Rotations

- Insert 55 to this AVL tree.
- Note that reducing the height of the pivot fixes that whole subtree!